NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

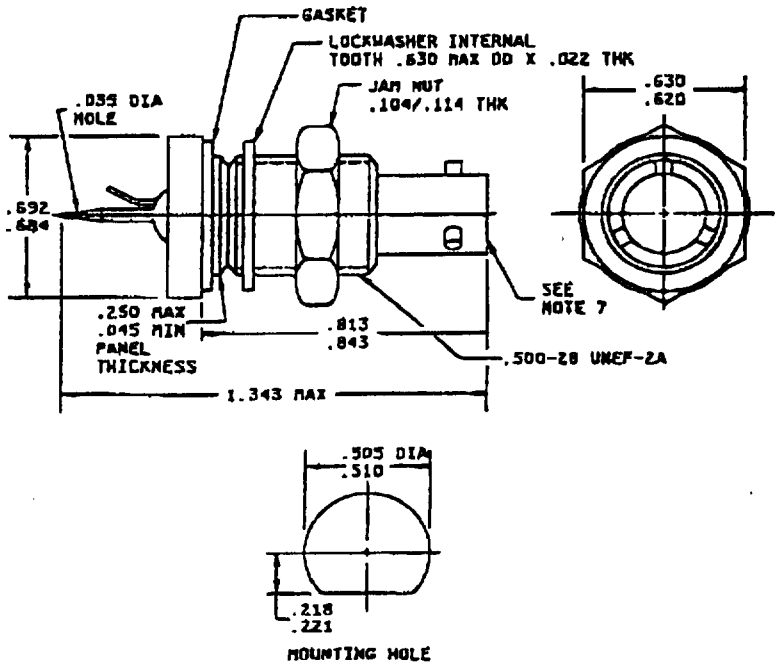
MIL-PRF-49142/4E
12 January 1990
SUPERSEDING
MIL-C-49142/4D
21 April 1988

PERFORMANCE SPECIFICATION

CONNECTORS, RECEPTACLE, ELECTRICAL, TRIAXIAL, RADIO FREQUENCY (SERIES TRB, SOCKET CONTACT, JAMNUT MOUNTED, CLASS 2) HERMETIC AND NONHERMETIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-49142



MOUNTING HOLE

Inches	mm
.022	.56
.035	.88
.045	1.14
.104	2.64
.114	2.90
.218	5.54
.221	5.61
.250	6.35
.500	12.70
.505	12.71
.510	12.95
.620	15.75
.630	16.00
.684	17.37
.692	17.58
.813	20.65
.843	21.41
1.343	34.11

NOTES:
1. Dimensions are in inches
2. Metric equivalents are given for general information only.
3. Wrench flats are to accommodate standard wrench openings per H-28, appendix 10.
4. All undimensioned pictorial representations are for reference purposes only.
5. Panel thickness, .045 (1.14 mm) minimum, .250 maximum.
6. Unless otherwise specified all tolerances are ±.005 inch.
7. Interface as specified in MIL-STD-348, series TRB, socket contact.

FIGURE 1. General configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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ENGINEERING PARAMETER·

Nominal impedance Nonconstant.

Frequency range· 0 to 500 MHz minimum

Voltage rating. 400 V rms, maximum working voltage at sea level. 100 V rms, maximum working voltage at 70,000 feet.

Temperature range -65°C to +165°C

REQUIREMENTS:

Dimensions and configuration: See figure 1 and MIL-STD-348.

Force to engage and disengage:
Longitudinal force: 4 pounds maximum.
Torque 2.5 inch-pounds maximum.

Coupling proof torque: Not applicable.

Mating characteristics: See figure 1 and MIL-STD-348 for dimensions.

Center contact (socket)
Oversize test pin 0 040 inch diameter minimum (nonclosed entry contacts only).
Insertion depth. .125 inch minimum.
Number of insertions 1

Insertion force test.
Steel test pin finish: 16 microinches.
Insertion force: 2 pounds maximum.
Steel test pin diameter· .039 minimum +.001 inch.

Withdrawal force test
Steel test pin diameter .037 maximum -.001 inch
Withdrawal force· 2 ounces minimum.
Test pin finish 16 microinches

Permeability: Applicable.

Hermetic seal: See table I.

Leakage· Connector shall be mounted in mounting hole shown on figure 1 with mating end capped. Air pressure 30 lb_f/in^2.

Insulation resistance 5000 megohms.

Conductor retention 6 pounds minimum, axial force.

Dielectric withstanding voltage: At sea level, 1200 V rms, between center conductor and intermediate conductor. 500 V rms, between intermediate conductor and outer conductor.

Corrosion (salt spray) Applicable.

Vibration Applicable.

Shock· Applicable

Thermal shock Applicable.

Moisture resistance Applicable

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Conductor resistance. In milliohms maximum.

	Connector dash number	Initial	After environment
Center conductor	0004 and all non-hermetic connectors	2.0	2.5
	0005 and all hermetic connectors	12	12.5
Intermediate conductor	0004 and all non-hermetic connectors	0.5	0.6
	0005 and all hermetic connectors	1.0	1.5
Outer conductor	0004 and all non-hermetic connectors	0.5	0.6
	0005 and all hermetic connectors	0.5	0.6

Corona level:
Altitude: 70,000 feet.
Voltage: 200 V rms minimum.

RF high potential withstanding voltage:
800 V rms, between center conductor and intermediate conductor.
200 V rms, between intermediate conductor and outer conductor at 5 to 7.5 MHz.

Leakage current: Not applicable.

Cable retention: Not applicable.

Coupling mechanism. Not applicable.

Rise time degradation: Not applicable.

Connector durability. 500 cycles, minimum, at 12 cycles per minute, maximum.

Part or Identifying Number (PIN): M49142/04- (dash number from table I.)

Not for Naval Air Systems Command (AS) use.

TABLE I. Cross reference of PIN.

Dash number	Superseded PIN
-0004 1/ -0104 -0204 -3004 1/ -3104 -3204	M49142/04-0001, -0002
-0005 1/ -0105 -0205 -3005 1/ -3105 -3205	M49142/04-0003

1/ Preferred keying.

Revision letters are not used to denote changes due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3750-01)

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